Università
degli Studi
di Padova

Messa a punto di un potenziale interatomico per la simulazione della tantala amorfa tramite machine-learning

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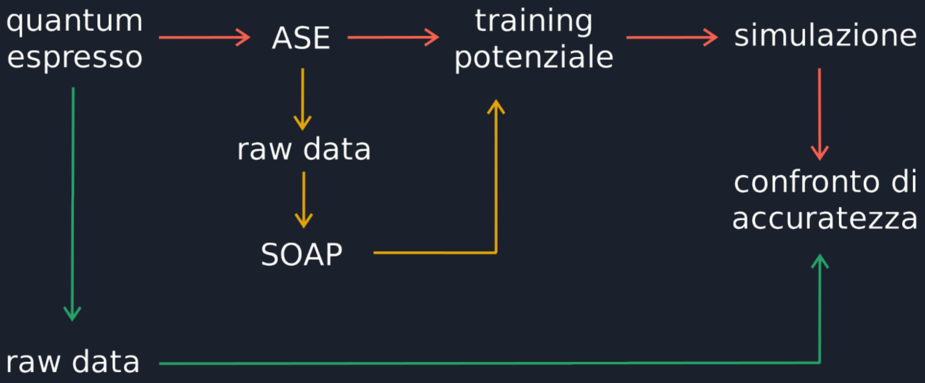
Struttura di lavoro

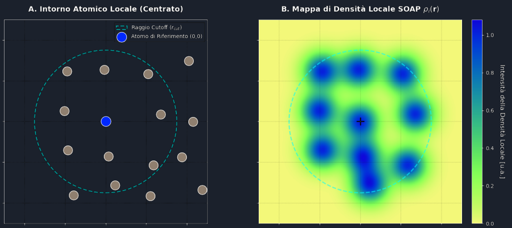


SOAP

SOAP: Smooth Overlap of Atomic Positions

Generazione della mappa di densità

$$\rho_Z(\vec{r}) = \sum_{\alpha}^{Z} e^{-\frac{|\vec{r}-\vec{r}_\alpha|^2}{2\sigma^2}} \tag{1}$$



SOAP

Espansione in funzioni di base

$$\rho_Z(\vec{r}) = \sum_{n=1}^{n_{max}} \sum_{l=1}^{l_{max}} \sum_{m=-l}^{l} c_Z(n,l,m) g(r\,;n) Y(\theta,\phi\,;l,m) \tag{2}$$

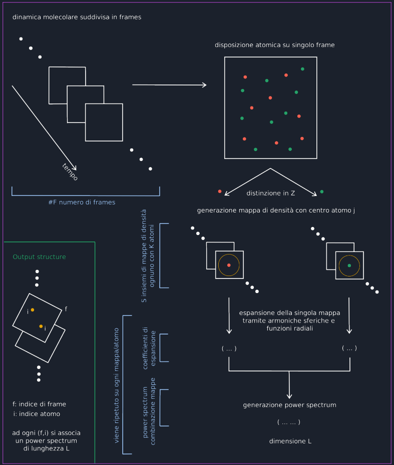
$$c_Z(n,l,m) = \int_{\mathbb{R}^3} \mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}\phi \;\; g(r\,;n) Y(\theta,\phi\,;l,m) \rho_Z(\vec{r}) \tag{3}$$

Calcolo del power spectrum

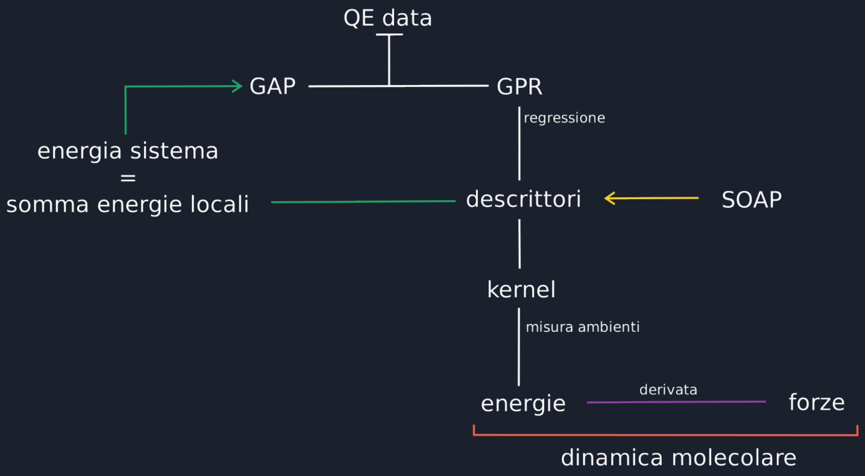
$$p_{Z_1 Z_2}(n,n',l) = \pi \sqrt{\frac{8}{2l+1}} \sum_{m=-l}^{l} \overline{c_{Z_1}}(n,l,m) c_{Z_2}(n',l,m) \tag{4}$$

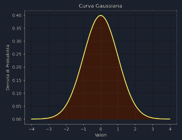
Risultato finale

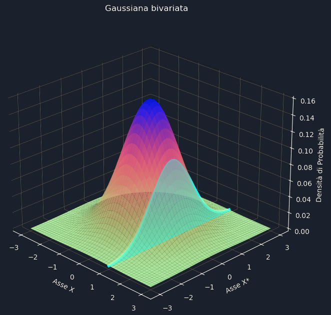
$$\dim = (l_{max}+1)\,\frac{1}{2}\,(S \cdot n_{max})(S \cdot n_{max}+1) \tag{5}$$



GAP


QE data
GAP
GPR
regressione
energia sistema
=
somma energie locali
descrittori
SOAP
kernel
misura ambienti
energie
derivata
forze
dinamica molecolare


Curva Gaussiana
Densità di Probabilità
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−4
−3
−2
−1
0
1
2
3
4
Valori


Gaussiana bivariata
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
Densità di Probabilità
Asse X
Asse X*
−3
−2
−1
0
1
2
3

Simulazione NVE

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ZBL: Potenziale Ziegler-Biersack-Littmark

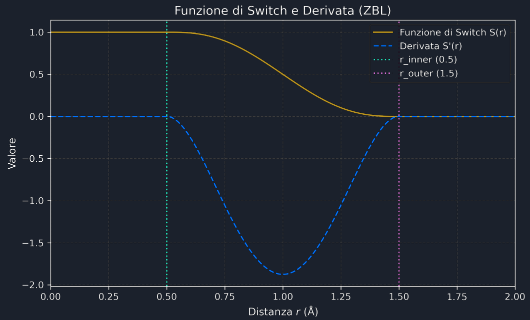
$$V(r_{ij}) = \frac{1}{4\pi\epsilon_0}\frac{Z_i Z_j}{r_{ij}}\phi\left(\frac{r_{ij}}{a_{ij}}\right) \tag{6}$$

$$\phi\left(\frac{r_{ij}}{a_{ij}}\right) = \phi(x) = 0.1818e^{-3.2x} + 0.5099e^{-0.9423x} + + 0.2802e^{-0.4029x} + 0.02817e^{-0.2016x} \tag{7}$$

$$a_{ij} = \frac{0.8854\, a_0}{Z_i^{0.23} + Z_j^{0.23}} \tag{8}$$

$$S(u) = -6u^5 + 15u^4 - 10u^3 + 1 \tag{9}$$

$$u = \frac{r - r_{inner}}{r_{outer} - r_{inner}} \tag{10}$$

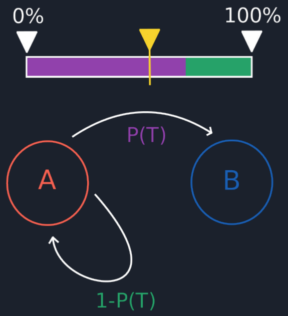


Simulazione con algoritmo di Metropolis

$$P(A) \propto \exp\left(-\frac{E_{pot}(A)}{k_b\,T}\right) \tag{11}$$

$$\Delta E = E_{new} - E_{old} \tag{12}$$

$$P_{acc} = \exp\left(-\frac{\Delta E}{k_b\,T}\right) \tag{13}$$



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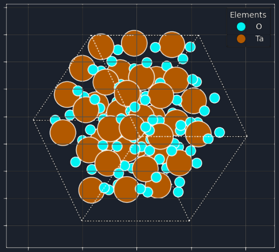
Dataset A



Verifica delle energie e delle forze

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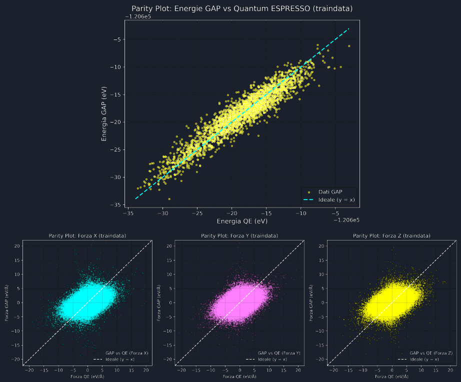
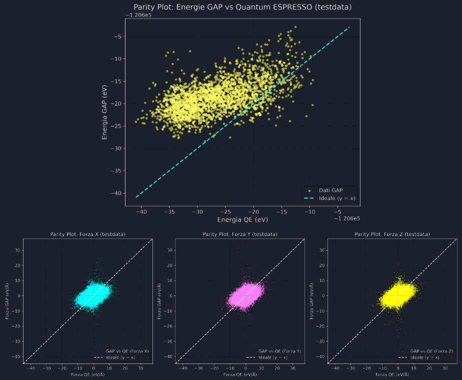
Training set



Test set



Simulazione NVE

Simulazione NVE: analisi

ZBL: risultati

Algoritmo di Metropolis: risultati

Domande??